

REDES NEURAIS COM TENSORFLOW



SÉRIES TEMPORAIS

DIEGO RODRIGUES DSC

INFNET

CRONOGRAMA

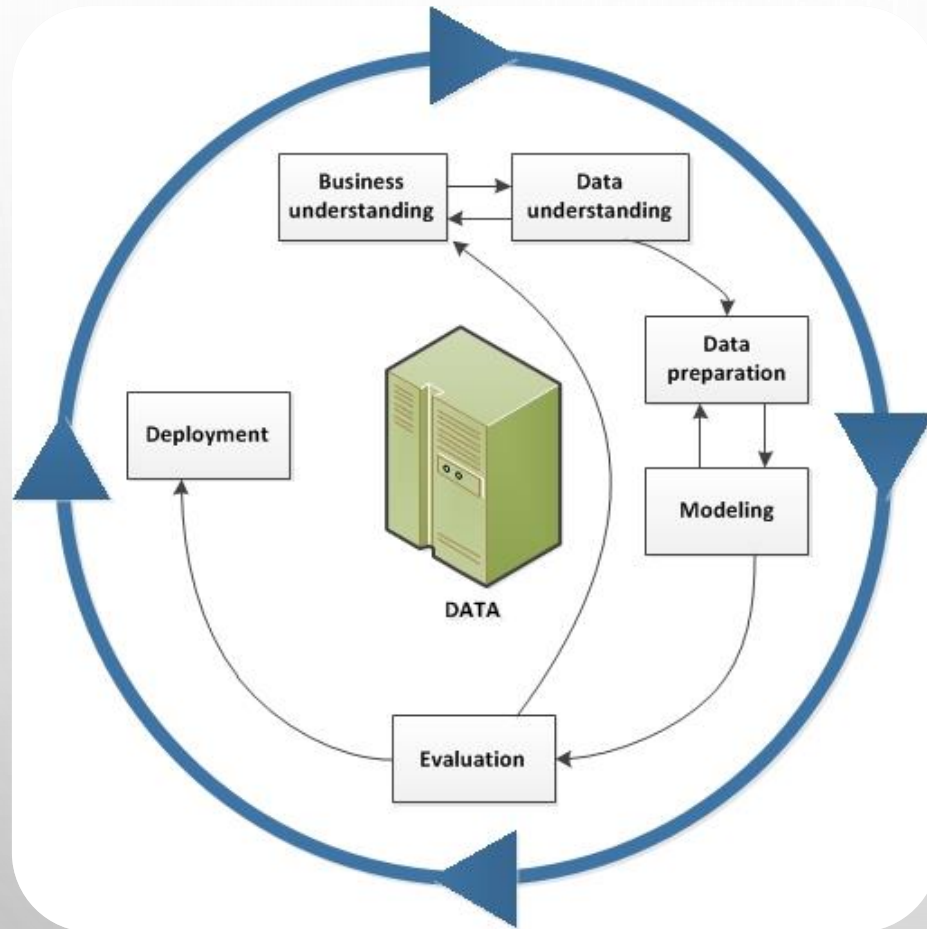
Dia	Aula	Trab
01	Perceptron de Rosenblatt	
02	Classificação: Neurônio Sigmóide	
03	Classificação: Rede Neural Feedforward	Grupos
04	Classificação: Rede Neural Profunda	
05	Regressão	Base de Dados
06	Agrupamento	
07	Séries Temporais	Modelos
08	Apresentação dos Trabalhos Parte I	



SÉRIES TEMPORAIS

- PARTE 1 : SÉRIES TEMPORAIS 101
 - BUSINESS UNDERSTANDING
 - SÉRIES TEMPORAIS
 - ABORDAGEM
 - DATA UNDERSTANDING & PREPARATION
 - ANÁLISE EXPLORATÓRIA
 - FEATURE EXTRACTION
 - MODELING
 - DECOMPOSIÇÃO
 - REGRESSÃO AUTO-VETOR
 - REDES NEURAS RECORRENTES
 - VALIDATION
 - SPLIT TREINO TESTE VALIDAÇÃO
- PARTE 2 : PRÁTICA
 - NOTEBOOK: AIR QUALITY TEMPERATURE
- PARTE 3 : TRABALHOS
 - ESCOPO & EVOLUÇÃO

PARTE 1 : TEORIA



CROSS INDUSTRY PROCESS FOR DATA MINING (CRISP-DM)

The background is a light gray gradient. In the top-left and bottom-right corners, there are several realistic water droplets of varying sizes, some overlapping. A faint, circular, embossed-like pattern is visible in the upper center of the page.

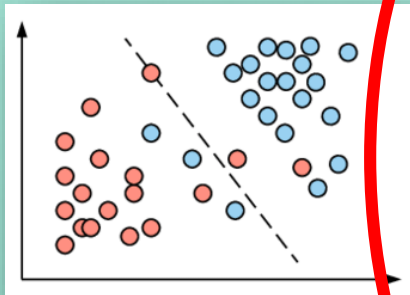
BUSINESS UNDERSTANDING

NOVO CICLO CRISP

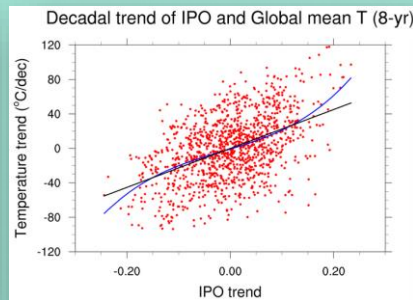
Algoritmo	Representação	Preparação	Modelagem	Validação
• Reta 2 Pontos • NN 10% VAL • NN 10 Folds • PS10 • PS10 • SOM • NN(T/V/T)	• 2D • 2D • 2D • 4D / 3 Classes • 7D / 1D Regressão • 4D • 1D 2 Lags	• Nenhuma • Nenhuma • Scale • Scale • Scale • PCA • Scale	• Reta 2 Pontos • 1 Neurônio • 1 Hidden • 1 Hidden • 1 Hidden • SOM 10x10 • 1 Hidden	• Nenhuma • Precisão/Recall • Precisão/Recall • Acurácia • MSE • MSE • MSE

- Regressão Séries Temporais com 2 Lags

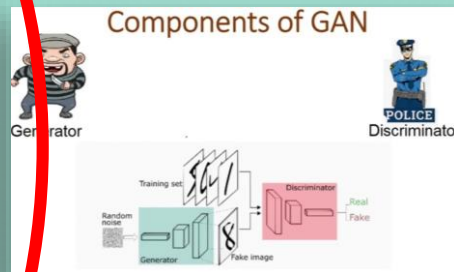
APRENDIZADO SUPERVISIONADO



CLASSIFICAÇÃO

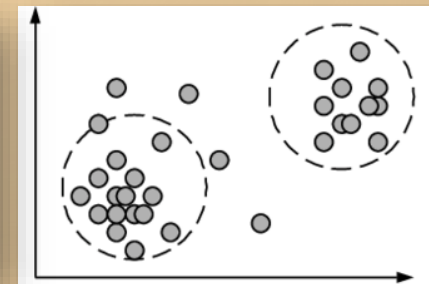


REGRESSÃO



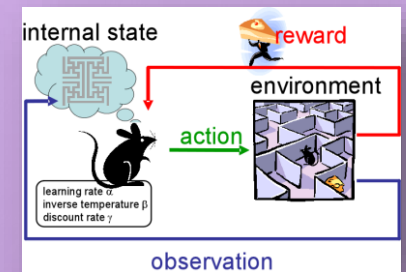
GENERATIVO

APRENDIZADO NÃO- SUPERVISIONADO



AGRUPAMENTO

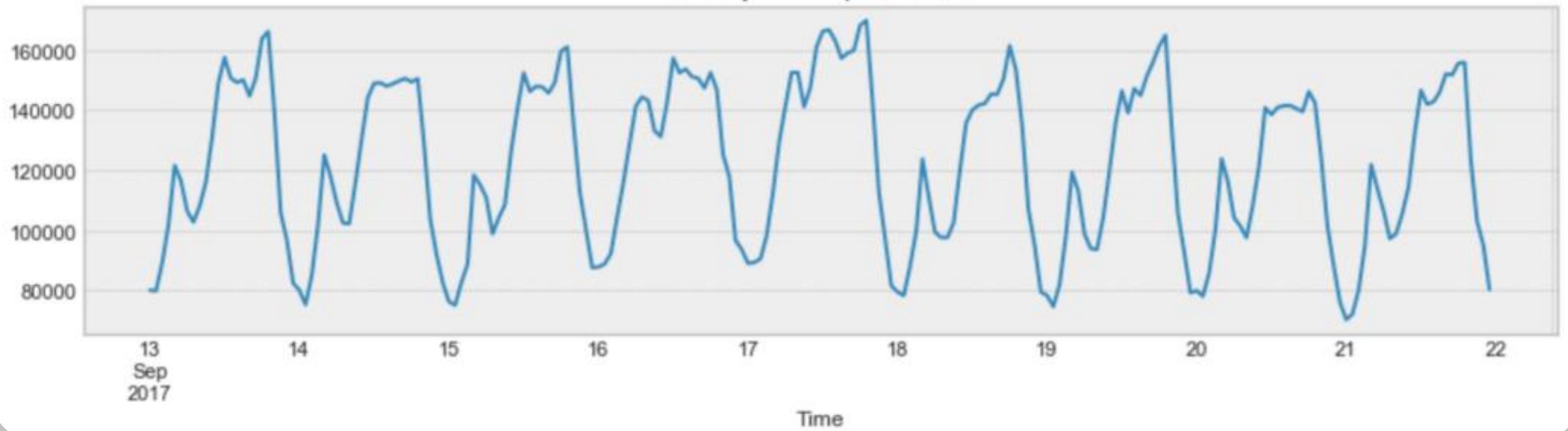
APRENDIZADO POR REFORÇO



REFORÇO

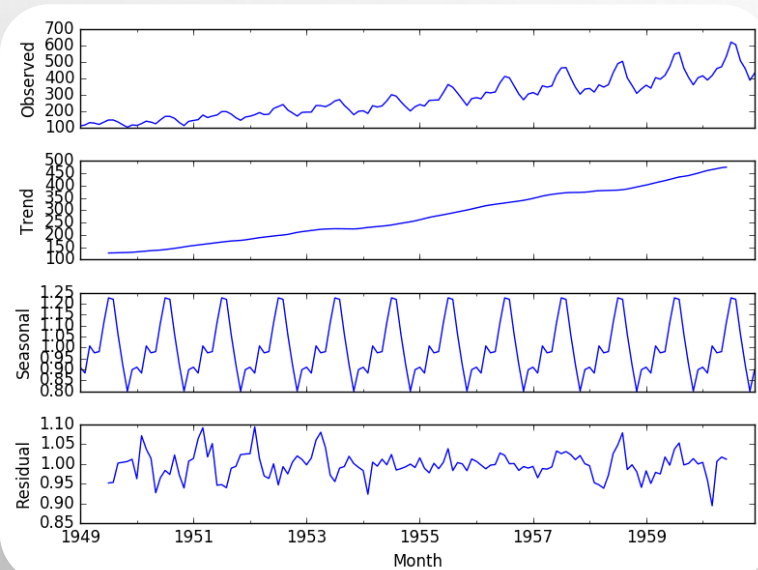
SÉRIES TEMPORAIS

Time Series Analysis Plots
Dickey-Fuller: $p=0.00000$

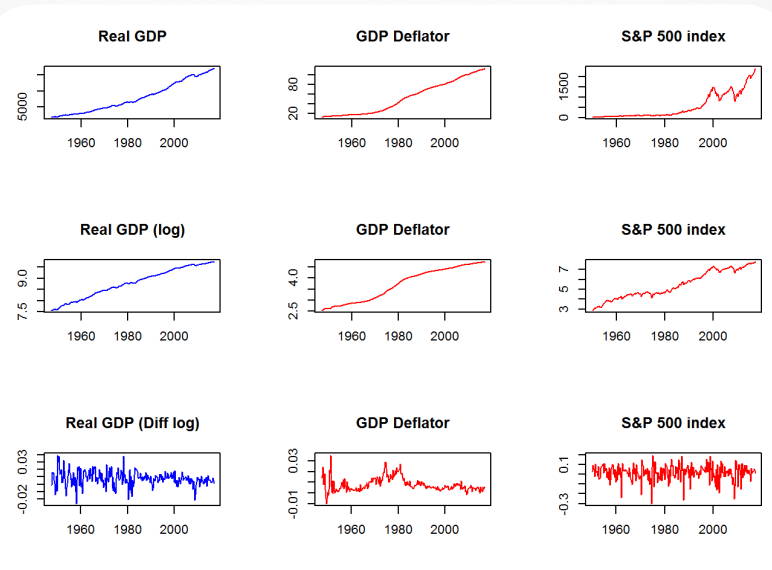


ABORDAGENS

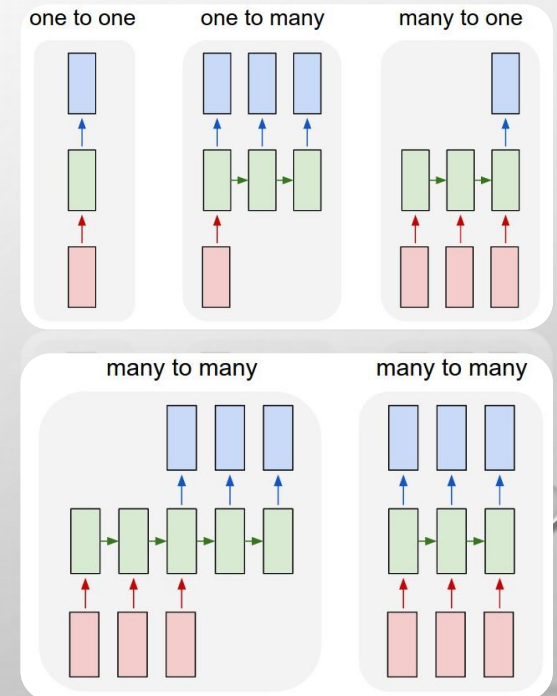
Decomposição



Regressão auto-vetor



Rede Neural Recorrente

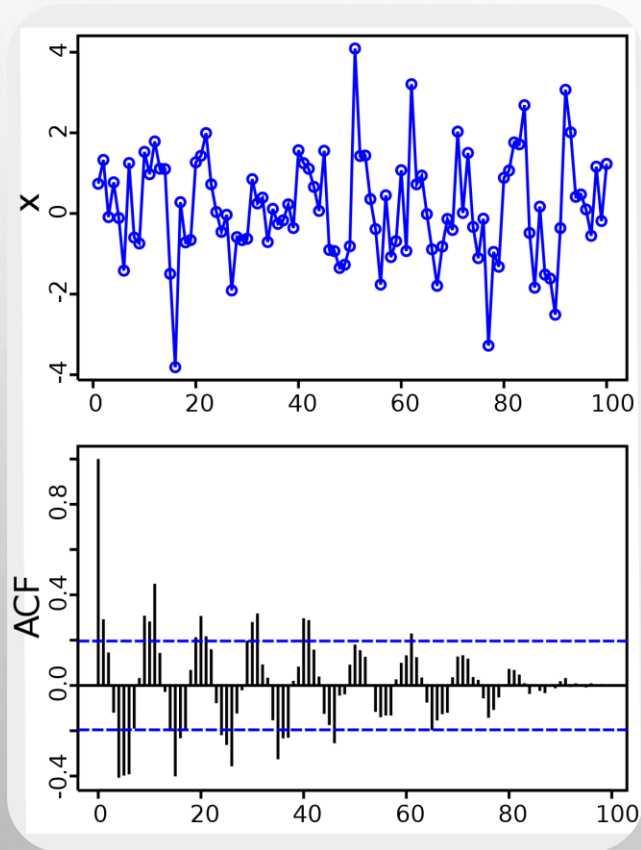


The slide features a light gray background with a subtle gradient. In the top-left and bottom-right corners, there are clusters of realistic water droplets of various sizes, some overlapping. A faint, circular, embossed-like pattern is visible in the upper center of the slide.

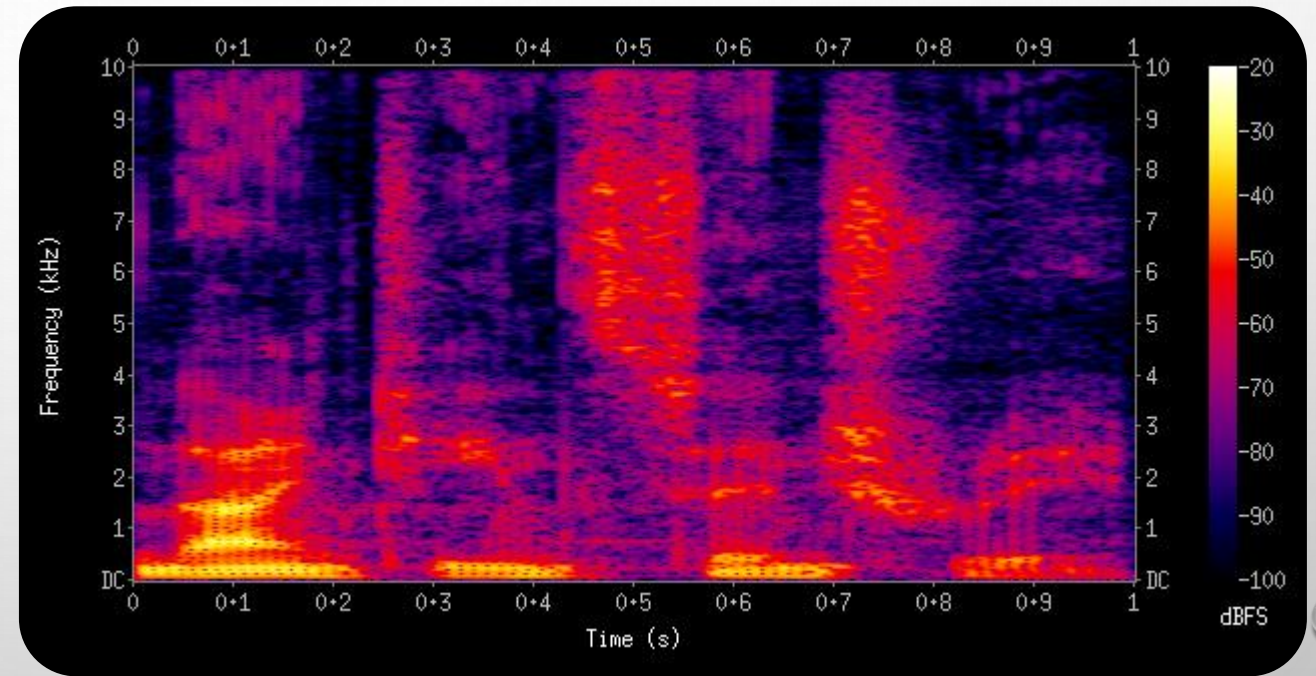
DATA PREPARATION & UNDERSTANDING

ANÁLISE EXPLORATÓRIA

Auto-correlação

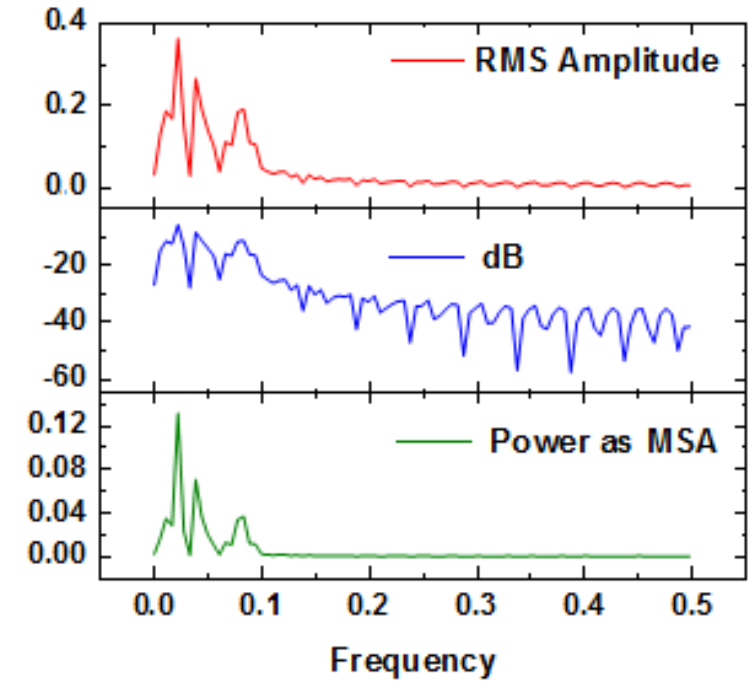
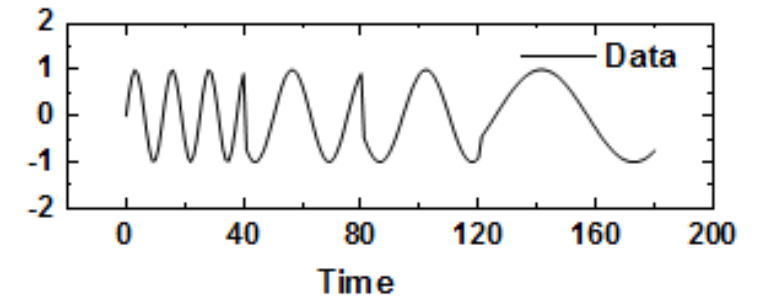
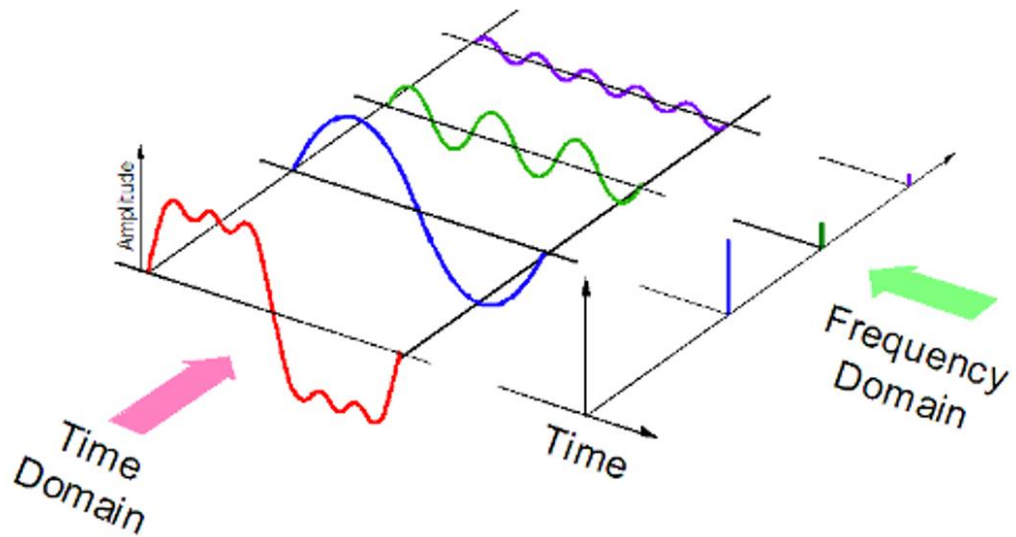


Espectrograma



FEATURE EXTRACTION

Transformada de Fourier



MODELING

SÉRIES TEMPORAIS

$$y_t = T_t + C_t + S_t + I_t,$$

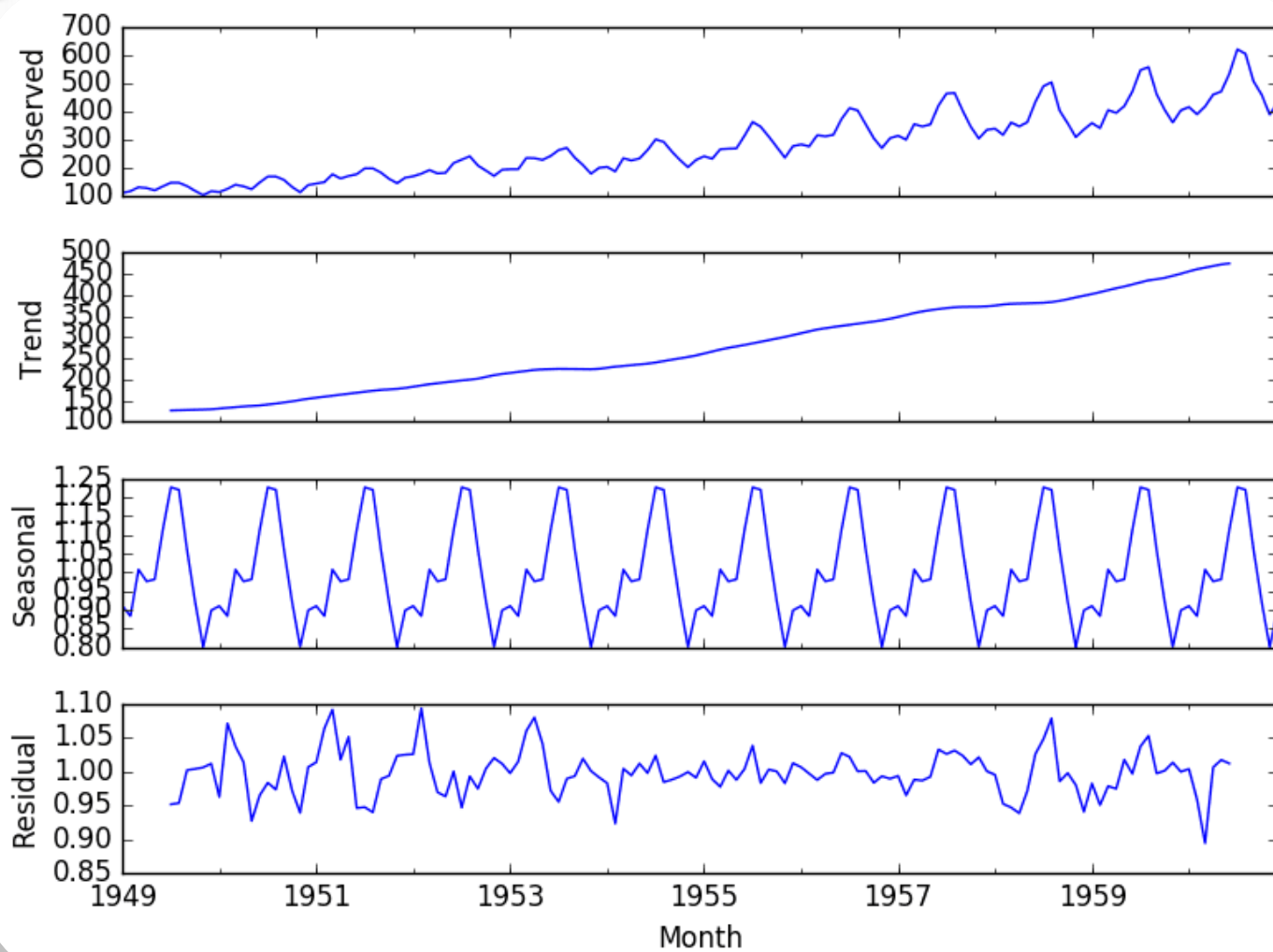
$$y_t = T_t \times C_t \times S_t \times I_t.$$

T_t : média / média móvel

C_t : Fourier Passa Baixas / Média Móvel

S_t : Fourier Passa Altas / Picos Remanescentes

I_t : auto-regressão / rede neural



DECOMPOSIÇÃO

REGRESSÃO AUTO-VETOR

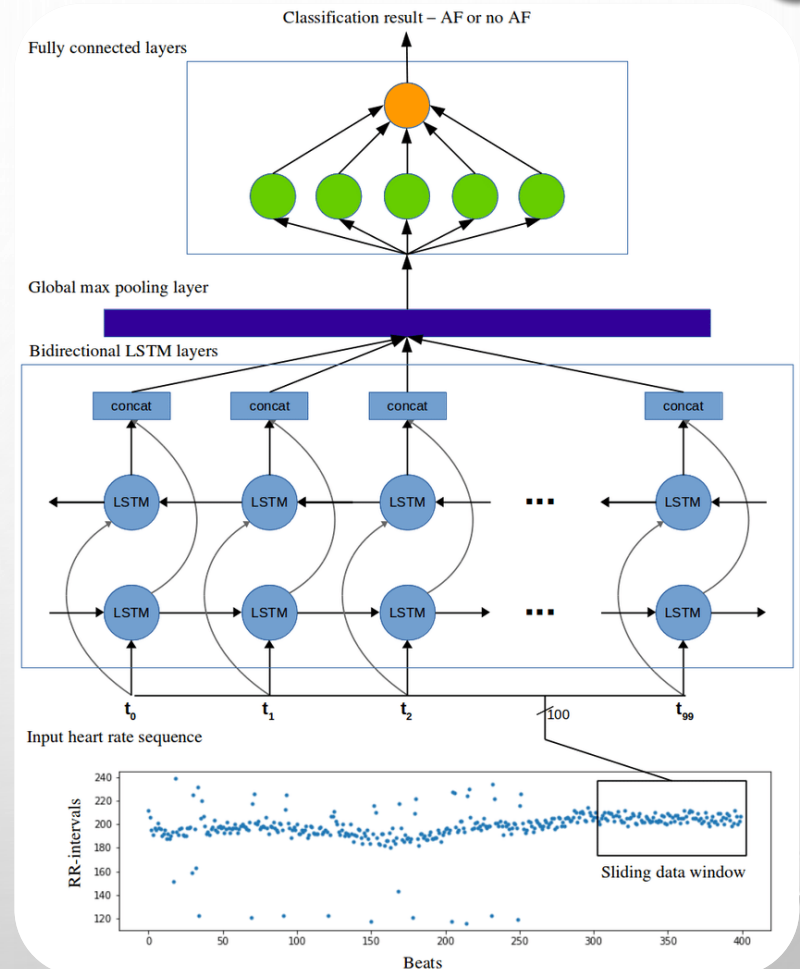
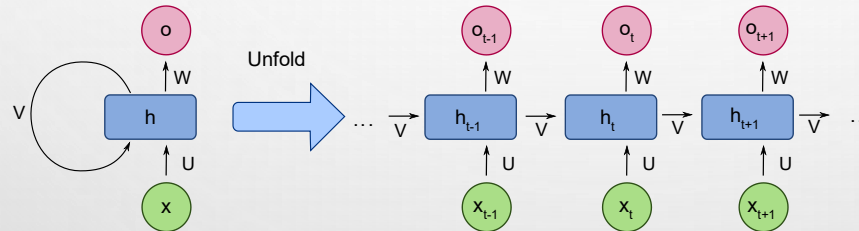
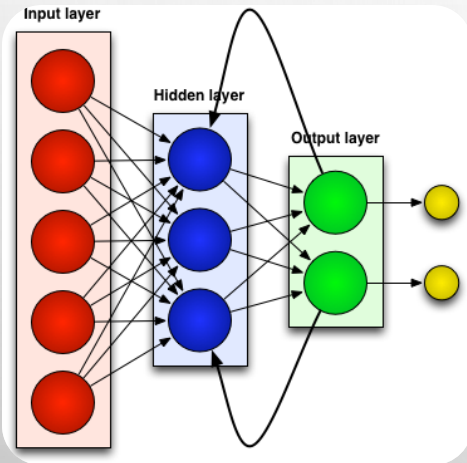
Univariada

$$X_t = c + \sum_{i=1}^p \varphi_i X_{t-i} + \varepsilon_t$$

Multivariada

$$y_t = c + A_1 y_{t-1} + A_2 y_{t-2} + \cdots + A_p y_{t-p} + e_t,$$

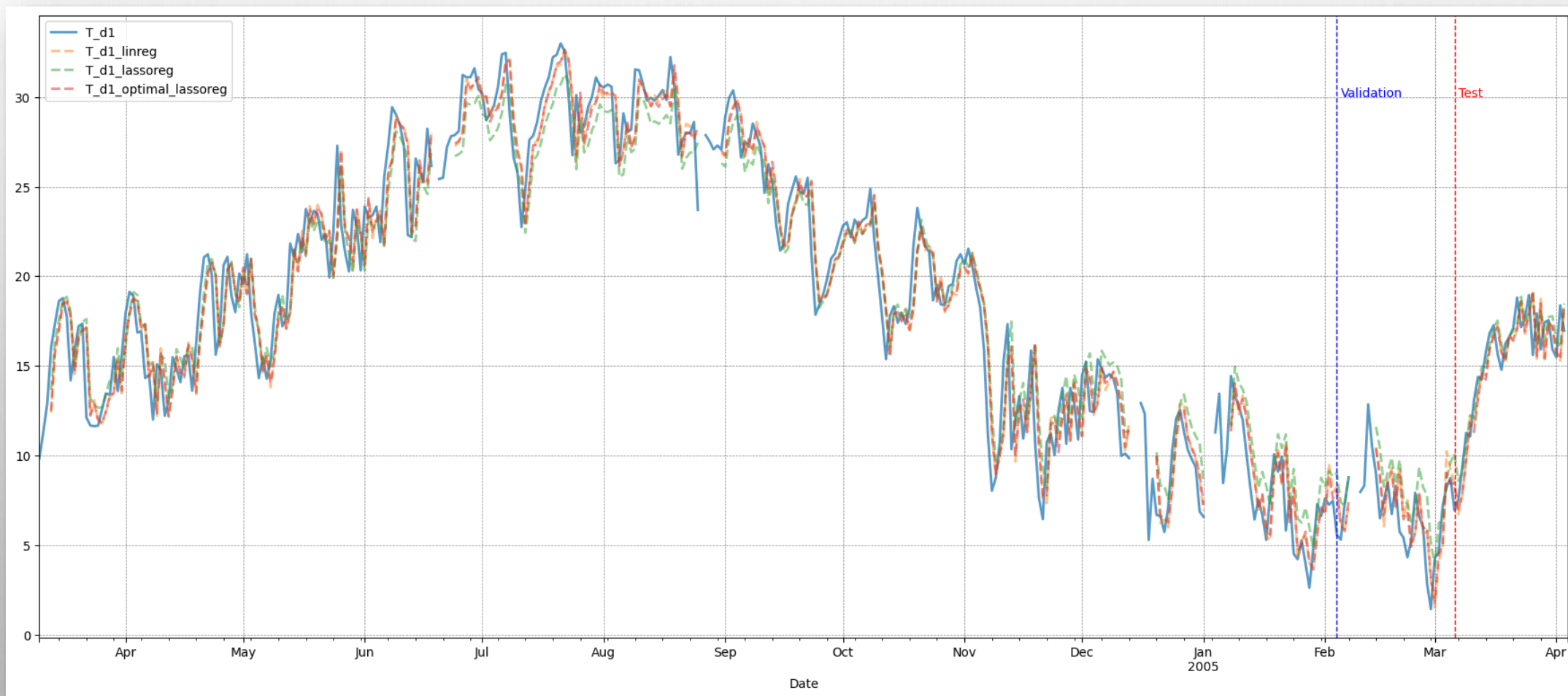
REDE NEURAL RECORRENTE





VALIDATION

SPLIT TREINO TESTE VALIDAÇÃO



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PARTE 2 : PRÁTICA

AMBIENTE PYTHON



4. Variáveis
Aleatórias

5. Visualização



6. Machine
Learning



1. Editor de Código



2. Gestor de Ambiente



3. Ambiente
Python do Projeto



3. Notebook
Dinâmico

PROBLEMA DE
NEGÓCIO

AirQuality Temperature

MODELAGEM

- **REDE NEURAL COM LAGS (AUTO-REGRESSÃO)**
 - 2 LAGS -> DIA SEGUINTE
 - 1 CAMADA OCULTA TANH
 - TREINAMENTO: (T/V/T)
 - MSE

**EXERCÍCIO: AIR QUALITY
REGRESSÃO LINEAR & REDE
NEURAL**

**PRÓXIMA AULA: APRESENTAÇÃO
PROPOSTAS DE TRABALHO**

